

Distill-and-Reinforce: Empowering BabyLM with Large Model Supervision

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Abstract

To bridge the gap between data-hungry LLMs and human-like data efficiency, we propose Distill-and-Reinforce for the BabyLM Challenge. This biologically plausible pipeline mimics development through three stages. (1) Teacher adaptation employs LoRA and Unsloth to adapt a GPT-2 Large teacher to child-directed speech while mitigating catastrophic forgetting. (2) Knowledge distillation transfers foundational linguistic competence to a GPT-2 Small student. (3) Interactive refinement optimizes the student through narrative generation and feedback from a Llama-3.1 teacher via PPO, GRPO, and DPO strategies. We observe that optimized teacher adaptation effectively preserves general reasoning. Furthermore, interactive feedback significantly enhances student performance, particularly in Entity Tracking, validating that combining domain-adapted distillation with social-like interaction is essential for data-efficient language learning.

1 Introduction

Recent advancements in Large Language Models (LLMs) have been predominantly driven by the “scaling hypothesis,” relying on trillions of training tokens to achieve high performance (Kaplan et al., 2020; Touvron et al., 2023). However, this data-hungry paradigm contrasts sharply with human cognitive development. Human children acquire fluent language and complex reasoning skills from a comparatively “poverty of the stimulus”—approximately 100 million words of exposure (Warstadt et al., 2023a). Crucially, human learning is not merely passive statistical absorption; it is deeply social, relying on interaction, correction, and feedback from “more knowledgeable others” such as parents or teachers (Vygotsky, 1978; Tomasello, 2003).

Motivated by this discrepancy, the BabyLM Challenge proposes a shift toward data-efficient

pretraining, limiting models to developmentally plausible data budgets. While recent works have explored architectural optimizations for small models (Eldan and Li, 2023), few have successfully mimicked the *interactive* nature of human learning to improve model performance under strict data constraints.

In this work, we propose a biologically plausible training pipeline for a small student model (GPT-2 Small) that mimics human development through two stages. First, we establish foundational language skill by passive knowledge distillation from a larger teacher model (GPT-2 Large). The teacher model is optimized by continual pretraining using LoRA and Unsloth to adapt to the domain of child-directed speech while reducing catastrophic forgetting. Second, we introduce an active learning phase where the student generates stories and receives feedback from a highly capable teacher (Llama-3.1-8B-Instruct). To optimize this interaction in the low-resource BabyLM setting, we evaluate three alignment strategies—Proximal Policy Optimization (PPO), Group Relative Policy Optimization (GRPO), and Direct Preference Optimization (DPO). Our results on the BabyLM 2025 evaluation suite demonstrate that combining domain-adapted distillation with interaction-based feedback significantly enhances reasoning capabilities, validating the hypothesis that interaction is essential for data-efficient language learning.

2 Related Work

2.1 Data-Efficient Language Modeling

Standard LLM training relies on massive corpora like CommonCrawl to saturate model parameters, often exceeding the token count a human encounters in a lifetime by orders of magnitude (Hoffmann et al., 2022). The BabyLM Challenge (Warstadt et al., 2023a) was established to investigate sample-efficient pretraining, restricting datasets to 10M or

100M words to mirror human childhood exposure. Recent approaches in this domain have focused on synthetic data generation, such as TinyStories (Elidan and Li, 2023), which demonstrated that small models (<100M parameters) can produce fluent English when trained on highly simplified, consistent synthetic text. However, while these models master syntax, they often lack the reasoning capabilities that emerge from larger-scale training or rich social interaction (Hu et al., 2023).

2.2 Knowledge Distillation (KD)

KD is a widely adopted technique where a small “student” model learns to mimic the behavior of a larger “teacher” (Hinton et al., 2015). In the context of NLP, this is typically achieved by minimizing the Kullback-Leibler (KL) divergence between the student’s and teacher’s output distributions (Sanh et al., 2019). For low-resource settings, standard logit-based KD can be insufficient. Advanced methods, such as Intermediate Layer Distillation (Jiao et al., 2020), align the hidden states of the student with the teacher, effectively transferring “reasoning patterns” rather than just final word probabilities. In our work, we utilize KD as a “scaffolding” mechanism to bootstrap the student’s capabilities before interactive fine-tuning.

2.3 Reinforcement Learning and Preference Optimization

Reinforcement Learning from Human Feedback (RLHF) has become the standard for aligning LLMs with user intent (Ouyang et al., 2022). The traditional approach utilizes Proximal Policy Optimization (PPO) (Schulman et al., 2017), an online RL algorithm that updates the policy to maximize a reward signal while constraining updates to ensure stability. While effective, PPO is computationally expensive and sensitive to hyperparameter tuning.

To mitigate these costs, Group Relative Policy Optimization (GRPO) (Shao et al., 2024) was introduced, which removes the need for a separate value function (critic) by normalizing rewards within a group of sampled outputs, significantly reducing memory overhead. Alternatively, offline methods like Direct Preference Optimization (DPO) (Rafailov et al., 2023) reparameterize the reward function to optimize the policy directly on preference pairs, bypassing the explicit reward modeling phase entirely. In this study, we utilize Llama-3.1 as an AI Judge (RLAIF) (Bai et al., 2022) to provide the feedback signal for these three algorithms.

2.4 Continual Pre-training

Standard large language models are typically trained on broad, general-domain corpora. However, for specialized tasks—such as the BabyLM challenge, which mimics early childhood language acquisition—direct application of these models may be suboptimal due to domain shifts between web-crawled text and child-directed speech. Continual pre-training addresses this by further training a pre-trained model on domain-specific data (Gururangan et al., 2020).

A major challenge in this process is *catastrophic forgetting*, where the model loses its previously acquired general knowledge while adapting to the new distribution (Kirkpatrick et al., 2017). To mitigate this, Parameter-Efficient Fine-Tuning (PEFT) methods, such as Low-Rank Adaptation (LoRA) (Hu et al., 2022), have been proposed. By freezing the pre-trained weights and optimizing only low-rank update matrices, LoRA allows the model to adapt to the BabyLM domain while retaining the general reasoning capabilities of the original “adult” model. Furthermore, we leverage **Unsloth** (Unsloth AI, 2024), an optimized training framework, to accelerate the continual pre-training process and reduce memory overhead effectively.

3 Modeling and Methodology

We propose *Distill-and-Reinforce*, a unified training pipeline designed to bridge the gap between large-scale reasoning capabilities and small-scale data constraints. As illustrated in Figure 1, our framework mimics the trajectory of human language acquisition through three consecutive stages:

- **Stage 1: Teacher Adaptation.** An “adult” teacher model adapts to the specific linguistic environment of a child.
- **Stage 2: Knowledge Distillation.** The teacher transfers its reasoning patterns to a smaller “child” student model via soft targets.
- **Stage 3: Interactive Refinement.** The student further refines its narrative generation capabilities through social interaction and reinforcement learning feedback from a teacher model.

In the following subsections, we detail the data setup and the specific implementation of each stage in this pipeline.

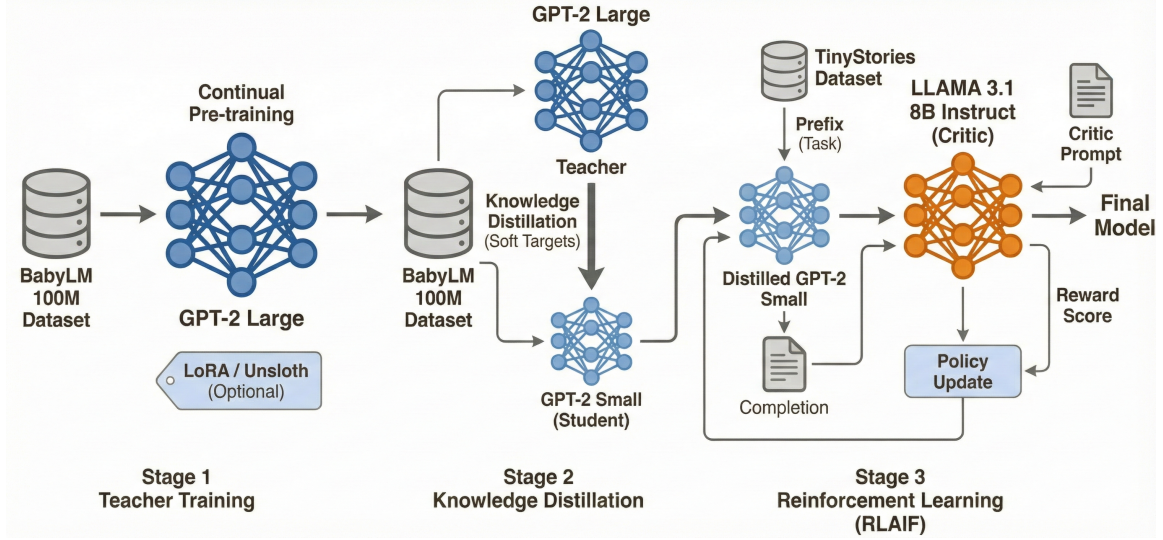


Figure 1: Overview of our *Distill-and-Reinforce* pipeline. The process flows from Domain Adaptation (Stage 1) to Knowledge Distillation (Stage 2), and finally to Interactive RL (Stage 3).

3.1 Data and Experimental Setup

We utilize the official datasets provided by the BabyLM Challenge (Charpentier et al., 2025). Unlike standard web-crawled corpora, this dataset is curated to simulate a **developmentally plausible** learning environment. It comprises diverse sources that a human child might encounter, including:

- **Child-Directed Speech:** Transcripts of spoken interactions between caregivers and children (derived from CHILDES).
- **Children’s Literature:** Text from simple stories and children’s books (e.g., CBT).
- **Simplified Knowledge:** Accessible reference materials such as Simple Wikipedia and simplified news articles.

For our experiments, we utilize two specific subsets to investigate scaling laws and supervision efficacy:

1. **BabyLM-10M (Strict-Small):** A strictly constrained subset of 10 million words. This serves as the primary training data for our main student baselines, mimicking the limited exposure of early childhood.
2. **BabyLM-100M:** A larger subset of 100 million words. We leverage this dataset for the continual pre-training of the teacher model (to ensure sufficient domain adaptation) and for training "strong" student variants to analyze the upper bounds of distillation performance.

All models utilize the same standard GPT-2 tokenizer to ensure vocabulary alignment.

3.2 Stage 1: Domain Adaptation via Continual Pre-training

The primary goal of the first stage is to adapt a general-purpose "adult" language model to the specific linguistic environment of a child. We select **GPT-2 Large** (774M parameters) as our base teacher model due to its strong reasoning capabilities.

Core Method: Continual Pre-training (CPT). We perform continual pre-training on the BabyLM-100M corpus. This process treats the domain adaptation as a standard causal language modeling task. Concretely, for each sequence of tokens $x_{1:T}$, we minimize the cross-entropy loss:

$$\mathcal{L}_{\text{LM}}^{\text{teacher}} = - \sum_{t=1}^T \log p_{\theta_T}(x_t \mid x_{<t}), \quad (1)$$

where θ_T denotes the teacher parameters. This approach allows the model to align its probability distributions with child-directed speech patterns.

Optimization: Mitigating Catastrophic Forgetting. While full parameter fine-tuning is effective for adaptation, it carries the risk of *catastrophic forgetting*, where the model loses general knowledge acquired during its original pre-training. To address this, we explore two optimization strategies:

- **LoRA (Low-Rank Adaptation):** Instead of updating all weights, we freeze the pre-trained

model and inject trainable rank decomposition matrices into the attention layers (Hu et al., 2022). This significantly reduces the number of trainable parameters while retaining the original knowledge structure.

- **Partial Unfreezing (Unsloth Method):** Standard LoRA freezes token embeddings, which limits the model’s ability to adapt to domain-specific vocabulary. Following the optimization strategy provided by **Unsloth (Unsloth AI, 2024)**, we unfreeze the embedding layer alongside LoRA but apply a significantly lower learning rate compared to other layers. This enables vocabulary adaptation while avoiding drastic drift leading to forgetting.

Teacher Variants. To evaluate the impact of these strategies, we implemented four distinct teacher configurations:

1. **Teacher-dft:** The off-the-shelf, pre-trained GPT-2 Large model without any adaptation. This serves as a zero-shot baseline.
2. **Teacher-1000M:** A GPT-2 Large fully fine-tuned (continual pre-trained) on the BabyLM-100M dataset for 10 epochs (equivalent to 1 billion tokens of exposure).
3. **Teacher-1000M-LoRA:** A model continually pre-trained for 10 epochs using LoRA, with all other parameters frozen.
4. **Teacher-1000M-LoRA-unfrozen:** A model continually pre-trained for 10 epochs using LoRA combined with unfrozen embedding layers (Unsloth optimization).

3.3 Stage 2: Knowledge Transfer via Distillation

In the second stage, we distill knowledge from the domain-adapted teacher T to a parameter-constrained student S (GPT-2 Small).

Principle: Soft Targets and KL Divergence. Standard training uses hard labels (one-hot vectors), which discard uncertainty. In contrast, our student learns from *soft targets*—the teacher’s output probabilities scaled by a temperature τ . Let z_i be the logit for the i -th class. The probability q_i is computed as:

$$q_i = \frac{\exp(z_i/\tau)}{\sum_j \exp(z_j/\tau)} \quad (2)$$

Higher τ produces a softer probability distribution, revealing rich structural information (e.g., semantic similarity) between classes. The student is trained to minimize a combined loss function:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{CE}} + (1 - \alpha) \tau^2 \mathcal{L}_{\text{KL}}(p_\tau^T || p_\tau^S) \quad (3)$$

where $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss with ground truth, $\mathcal{L}_{\text{KL}}$ is the Kullback-Leibler divergence between the teacher’s soft targets p_τ^T and the student’s predictions p_τ^S , and α balances the two terms.

Student Variants. To rigorously analyze the impact of teacher quality and training duration, we compare multiple student configurations. Note that in our naming convention (e.g., *Stu-900M*), the suffix denotes the total number of tokens seen during training, not necessarily the unique dataset size.

1. Baselines (No Distillation):

- **BabyLM-10M / BabyLM-100M:** GPT-2 Small models trained from scratch on the 10M or 100M datasets using only standard cross-entropy loss ($\mathcal{L}_{\text{CE}}$).

2. Students of Default Teacher (Teacher-dft):

To isolate the gains provided by Stage 1 adaptation, we distill from the non-adapted teacher:

- **Stu-100M-dft:** Distilled on the 10M dataset for 10 epochs (Total 100M tokens).
- **Stu-900M-dft:** Distilled on the 100M dataset for 9 epochs (Total 900M tokens).

3. Students of Adapted Teacher (Teacher-1000M):

These students are distilled from the standard continually pre-trained teacher:

- **Stu-10M:** Distilled on the 10M dataset for 1 epoch. This is our primary configuration for the Strict-Small track.
- **Stu-100M:** Distilled on the 10M dataset for 10 epochs.
- **Stu-900M:** Distilled on the 100M dataset for 9 epochs.

4. Students of Optimized Teachers:

To evaluate if the LoRA/Unsloth optimization in Stage 1 benefits the student downstream:

- **Stu-900M-LoRA:** Distilled from *Teacher-1000M-LoRA* on the 100M dataset (9 epochs).

- **Stu-900M-LoRA-unfrozen:** Distilled from *Teacher-1000M-LoRA-unfrozen* on the 100M dataset (9 epochs).

3.4 Stage 3: Reinforcement Learning Interactive Improvement

In the final stage, the student model transitions from passive supervised learning to an active interactive learning paradigm. The objective is to refine the student’s generation capabilities by simulating a social learning environment where the model generates narrative outputs and receives scalar feedback from a highly capable adult teacher model.

Formally, given a prompt x sourced from the TinyStories dataset, the student policy π_θ generates a completion y . A reward model (the teacher) provides a scalar reward $R(x, y)$, which serves as the reinforcement signal to update the policy parameters θ . We employ **Llama-3.1-8B-Instruct** as the teacher model and investigate three distinct alignment algorithms: PPO, GRPO, and DPO.

Reward Formulation The feedback signal is derived from a holistic evaluation of the generated story. We query the teacher model to score the completion y on three qualitative dimensions: Readability (S_{read}), Narrative Coherence (S_{narr}), and Creativity (S_{creat}).

To mitigate the tendency of smaller models to produce premature or truncated endings, we introduce a length penalty term. The final reward R is computed as a weighted combination of the teacher’s average qualitative assessment and a length bonus:

$$R(x, y) = \frac{\bar{S}_{qual} + \lambda \cdot \mathbf{1}(|y| > L_{min})}{1 + \lambda} \quad (4)$$

where $\bar{S}_{qual} = \frac{1}{3}(S_{read} + S_{narr} + S_{creat})$ is the mean of the qualitative scores, $\mathbf{1}$ is the indicator function, L_{min} is the length threshold, and λ is a balancing coefficient that controls the weight of the length constraint relative to quality.

Proximal Policy Optimization (PPO) PPO (Schulman et al., 2017) serves as our baseline for online reinforcement learning. PPO operates under an Actor-Critic architecture, requiring two distinct networks: the policy π_θ (Actor) and a value function V_ϕ (Critic).

The Critic is trained to estimate the expected return of a given state. During training, the advantage

A_t is estimated as the discrepancy between the actual realized reward R and the Critic’s prediction:

$$A_t = R(x, y) - V_\phi(x, y_{<t}) \quad (5)$$

The policy is updated to maximize the expected reward while constraining the update size via a clipped objective function to ensure stability. Ideally, the student learns to exceed expectations by generating outputs that yield higher rewards than predicted. However, we observed that training a stable Critic on a small-scale model (BabyLM-100M parameters) was computationally expensive and prone to high variance.

Group Relative Policy Optimization (GRPO)

To address the instability associated with the Critic network, we implement GRPO (Shao et al., 2024). GRPO eliminates the value network entirely, relying instead on group-based variance reduction.

For each prompt x , the student policy generates a group of G outputs $\{y_1, y_2, \dots, y_G\}$. The advantage for the i -th output is calculated by normalizing its reward against the statistics of the group:

$$A_i = \frac{R(x, y_i) - \mu_{group}}{\sigma_{group} + \epsilon} \quad (6)$$

where μ_{group} and σ_{group} are the mean and standard deviation of the rewards within the generated batch. By using the group average as a dynamic baseline, the student effectively competes against its own performance level. This method significantly reduces memory overhead by removing the need for separate value model parameters.

Direct Preference Optimization (DPO) Finally, we explore DPO (Rafailov et al., 2023), which reparameterizes the RL objective as a classification problem, bypassing explicit reward modeling during the update phase.

Although typically an offline method, we implement it in an online fashion. For each prompt x , the student generates a pair of completions (y_1, y_2) . The teacher scores both to construct a preference pair (y_w, y_l) , where y_w is the completion with the higher reward (winner) and y_l is the lower (loser). The student is optimized to minimize the negative log-likelihood of the preferred completion relative to the rejected one:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right] \quad (7)$$

where π_{ref} is the frozen reference model (the student state before the current RL round) and β controls the strength of the divergence penalty. This approach directly optimizes the probability margin between superior and inferior narratives.

4 Experiments and Results

4.1 Zero-shot Evaluation Benchmarks

The evaluation pipeline partly follows the 2025 BabyLM Challenge (Charpentier et al., 2025). We evaluate the models’ general semantic abilities and systematic generalization capabilities with the following benchmarks provided in the sentence zero-shot section: BLiMP, BLiMP supplement, EWoK, COMPS, WUGs, and entity tracking (ET). Compared to the standard zero-shot evaluation pipeline of the 2025 BabyLM Challenge, here we skip eye-tracking (Eye-T) and self-paced reading (SPR), which aim to evaluate human-reading likeness of the model when dealing with long textual inputs.

The zero-shot evaluation benchmarks consist of the following tasks:

- BLiMP (The Benchmark of Linguistic Minimal Pairs) (Warstadt et al., 2023b) evaluates the grammatical acceptability of language models by checking whether they assign a higher probability to the grammatical sentence in each test pair.
- BLiMP Supplement covers more comprehensive test pairs to give more robust evaluation compared to the basic BLiMP task.
- EWoK (Elements of World Knowledge) (Ivanova et al., 2025) evaluates the language models on specific world concepts such as social interactions, spatial relations, and physical dynamics.
- COMPS (Conceptual Minimal Pair Sentences) (Misra et al., 2023) targets on conceptual and semantic abilities of the language models using minimal pairs.
- ET (Entity Tracking) (Kim and Schuster, 2023) evaluate the models’ reasoning abilities by checking whether they can maintain reference to entities across discourse.
- WAG-A (Hofmann et al., 2024) tests morphological generalization in language models through

an adjective nominalization (-ity vs. -ness). task.

- WAG-P (Weissweiler et al., 2023) tests morphological generalization by evaluating the model-predicted past tense forms of nonce words.

4.2 Baselines

The baseline models include:

SimPO: The baseline model provided by BabyLM Challenge 2025 interaction track, where the student model is first trained with the language modeling loss on the full teacher-corrected datapoint and is then further finetuned with a preference optimization algorithm SimPO (Meng et al., 2024).

BLM: The state-of-art and the winner of BabyLM Challenge 2025 interaction track (Martins et al., 2025).

4.3 Results and Analysis

Compare to the baselines. Table 1 shows the zero-shot evaluation results of our models and the baselines. The corresponding bar chart is provided in Figure 2. Compared to the model trained from scratch (BabyLM-1000M), our models outperform the baseline across all the tasks, demonstrating the effectiveness of knowledge distillation. However, the size of the training corpus constrains the performance of the student models (Stu-100M vs. Stu-900M). When trained on a larger corpus (100M tokens), the student model performs better on all the tasks, with significant improvement on ET. Compared to the baseline SimPO and the state-of-the-art BLM, both Stu-900M and Stu-900M-PPO-1M outperform or match performance on most tasks except for WUG-P, where SimPO performs significantly better. Thus, our distill-and-reinforce strategy achieves success in this data-constrained BabyLM task.

Reinforcement Learning. Following the training pipeline in BLM, first we apply PPO as the RL algorithm. In Figure 2 we observe that after interactive RL, the student models show improvement in ET. This indicates that RL can help the student model maintain contextual reference and improve discourse coherence. However, the performance of the RL student model decreases on several tasks, such as BLiMP, EWoK, and WUG-A. Thus, we explore DPO and GRPO as alternative RL algorithms, as introduced in section 3.4. Evaluation results of the RL student models with different RL

Task	SimPO	BLM	Teacher-1000M	BabyLM-1000M	Stu-100M	Stu-100M-PPO-1M	Stu-900M	Stu-900M-PPO-1M
BLiMP	72.16	77.53	81.15	60.00	73.66	75.86	<u>77.97</u>	77.30
Suppl.	61.22	56.72	72.19	56.83	61.74	62.06	63.44	61.48
EWoK	51.92	51.41	59.22	50.7	51.81	51.05	<u>53.47</u>	52.67
COMPS	55.05	55.18	64.70	49.93	56.33	56.64	<u>59.40</u>	59.20
ET	28.06	<u>33.11</u>	40.76	23.46	17.42	16.64	31.77	32.53
WUG-A	0.510	0.711	0.56	0.42	0.59	0.55	<u>0.67</u>	0.63
WUG-P	<u>0.179</u>	0.045	0.22	-0.26	-0.05	-0.02	0.06	0.06

Table 1: Evaluation results for BabyLM models. **Teacher-1000M**: Continual pre-trained GPT2-large model on 100M corpus. **BabyLM-1000M**: GPT2-small baseline trained from scratch on 100M corpus. **Stu-100M**: GPT2-small distilled from teacher on 10M corpus. **Stu-900M**: GPT2-small distilled from teacher on 100M corpus. **Stu-100M-PPO-1M & Stu-900M-PPO-1M**: Further trained student models using PPO with 1M generation budget. **SimPO & BLM**: Interaction track baselines. **Bold**: The best result; Underlined: The second best.

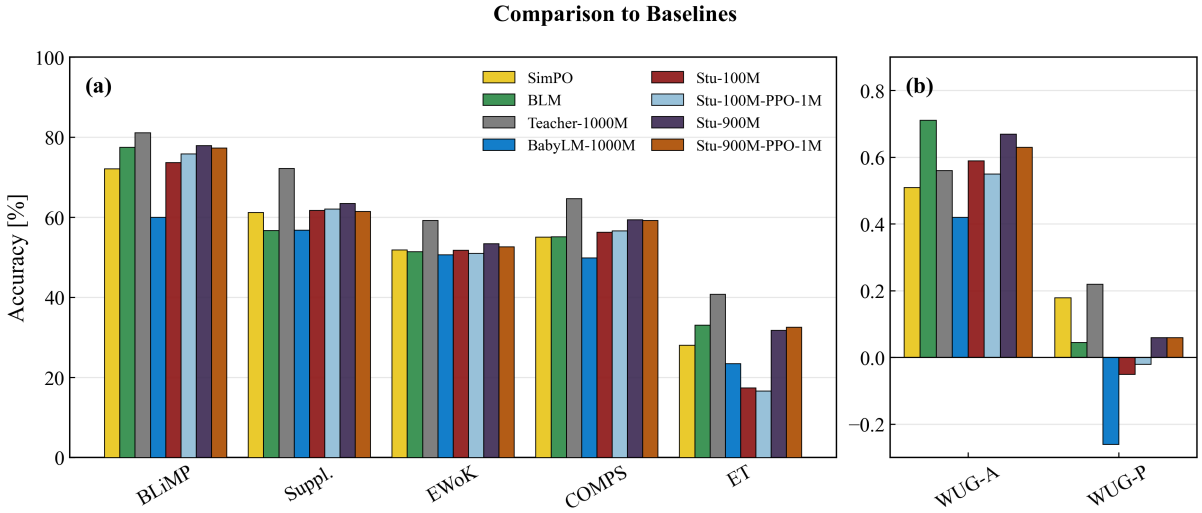


Figure 2: Bar chart of the evaluation results for BabyLM models.

algorithms are listed in Table 2, with the respective bar chart in Figure 3. Compared to the original PPO, DPO shows marginal improvements on BLiMP, BLiMP Supplement, EWoK, and WUG-A. GRPO achieves significant improvements on ET, WUG-A, and WUG-P tasks. Combined with an inspection of storytelling quality (refer to A. Appendix), GRPO model produces more sensible and coherent story generations.

Optimized continual pre-training on the teacher model.

During the KD stage, the student model inherits the teacher model’s performance, indicating that enhancing the performance of teacher model is crucial for a more robust student model. Although domain knowledge adaptation on teacher model is important in this project, this continual pre-training can harm the model’s performance and even lead to catastrophic forgetting. This is evi-

denced by the performance gaps between Teacher-1000M in Table 1 and Teacher-dft in Table 3, which are the continual pre-trained teacher and the default GPT2-large model without continual trained, respectively. To mitigate this, we apply optimized continual pre-trained methods as introduced in section 3.2. Evaluation results are shown in Table 3 and Figure 4.

Notably, the naive continual pre-training method leads to performance decays mainly on BLiMP, BLiMP Supplement, EWoK, ET and WUGs. LoRA helps mitigate forgetting on BLiMP, ET and WUGs, while combining LoRA with embedding layer unfrozen further improves the teacher model’s performance on BLiMP Supplement and the student model’s performance on ET.

Task	Stu-900M-GRPO-1M	Stu-900M-DPO-1M	Stu-900M-PPO-1M
BLiMP	76.54	77.86	77.30
Suppl.	60.53	63.23	61.48
EWoK	52.67	53.83	52.67
COMPS	57.34	57.83	59.20
ET	38.99	31.80	32.53
WUG-A	0.71	0.65	0.63
WUG-P	0.15	0.05	0.06

Table 2: Evaluation on the models trained using different RL algorithms

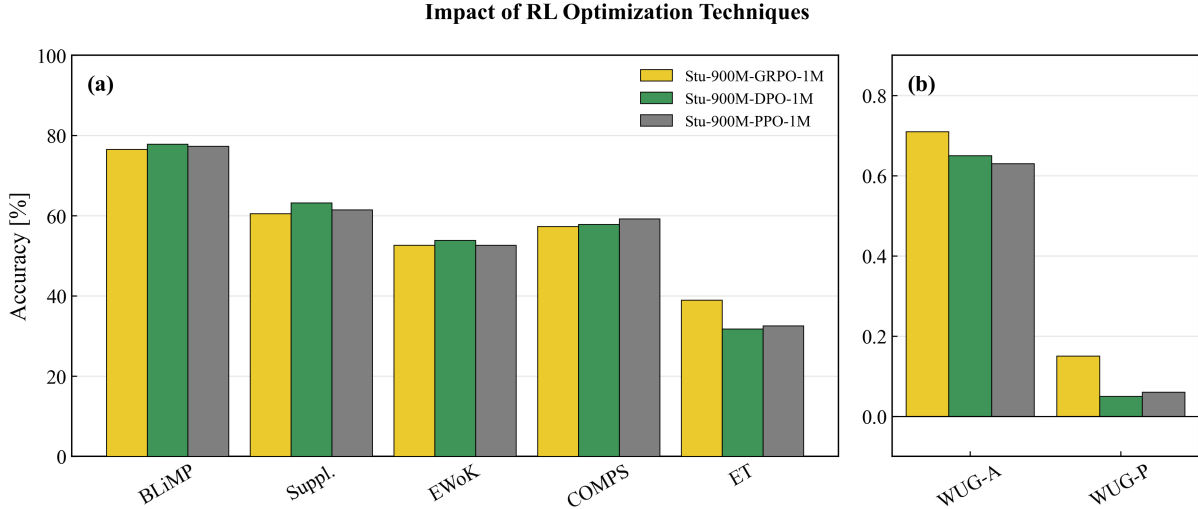


Figure 3: Bar chart of the evaluation results for optimized RL models.

5 Conclusion

Our experiments demonstrate that the distill-and-reinforce strategy is highly efficient for this data-constrained BabyLM domain. Our models achieve expected performance such that outperform the SimPO and BLM baselines on most tasks. We also explore the impact of different RL algorithms and the optimized continual pre-training methods to boost student performance.

We find that the first stage continual pre-training on the teacher model is crucial, and the naive continual pre-training method causes catastrophic forgetting issues. Optimized continual pre-training techniques, including LoRA and embedding layer unfrozen, effectively maintain the teacher’s general abilities during domain knowledge adaptation. Knowledge distillation allows the student model to acquire domain knowledge more efficiently than training from scratch, with student performance closely correlating to that of the teacher. The third stage interactive RL enhances narrative coherence and contextual reference maintenance abilities—specifically entity tracking—while leaving

the other tasks largely unchanged. This improvement in entity tracking aligns with our storytelling goals, and combined with GRPO algorithm, the entity tracking ability is further enhanced.

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Task	Teacher-dft	Stu-900M-dft	Teacher-LoRA	Stu-900M-LoRA	Teacher-LoRA-unfrozen	Stu-900M-LoRA-unfrozen
BLiMP	83.24	79.16	82.52	78.36	83.00	79.67
Suppl.	73.51	64.71	72.07	64.11	74.88	65.63
EWoK	61.73	53.19	58.65	53.18	58.64	53.75
COMPS	63.27	59.00	65.12	59.33	65.04	60.01
ET	45.22	34.18	45.98	20.18	44.84	29.20
WUG-A	0.60	0.69	0.57	0.66	0.55	0.65
WUG-P	0.24	-0.02	0.27	0.01	0.27	0.01

Table 3: Evaluation results on the default teacher-student models and the optimized continual pre-trained teacher-student models.

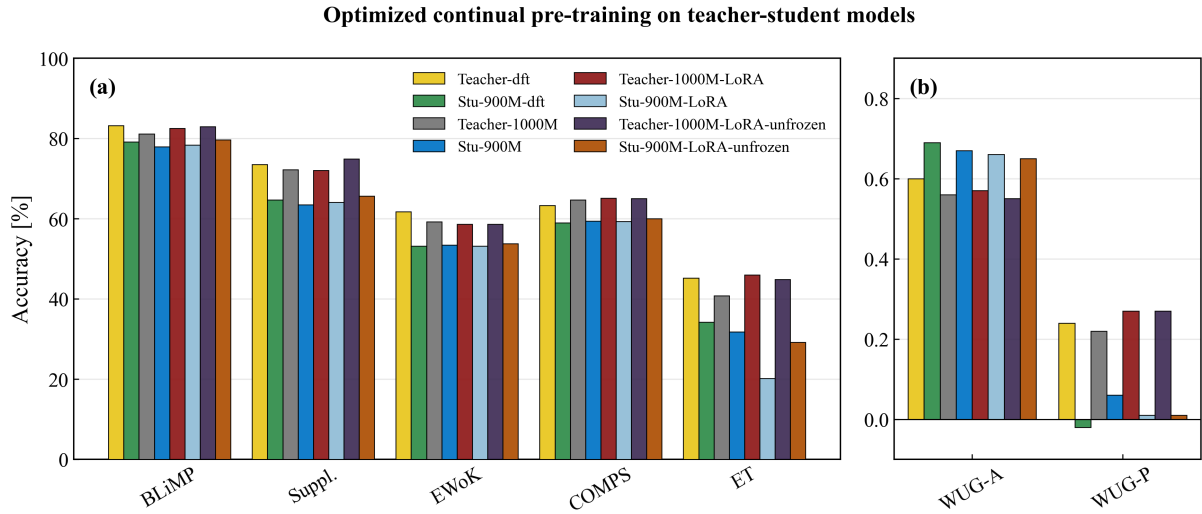


Figure 4: Bar chart of the evaluation results for optimized continual pre-trained models.

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A Appendix

A.1 Story Example

To illustrate the qualitative differences between the alignment strategies, we present sample completions for the prompt **"Once upon a time, in"** generated during the final stages of training.

PPO It updates the policy based on a single generation per step compared against a value prediction. The following examples are from **step 3865** and **step 4068**.

Student: " Dewey’s wheel-house, who had set his Spaniards in the trap. He, riding away from them, dismounted a stone, stooped, crept out of their way, and faced his nieces. Give him one moment, said one; the mission is over. the huntsman only laughed, and said, He must have been his brazen master, after all. the huntsman answered"

Score: 0.89

Student: " the morning, then, on a barrel, riding his frail and daft back, close beside his astonished mistress Mary and Belle, stood her little doll herself perched upright on the two wooden stone steps of the room, with the band propped up under her

little neck and the hands smoothed back and forth to keep the cold wind in her ears. I am worn out with wishing you to come back again, Hannah! she"

Score: 0.78

GRPO In GRPO, a group of students (e.g., $G = 4$) generate completions simultaneously for the same prompt. The model updates based on the relative performance within this group. The following examples are from **step 59467**, where the model has largely converged, resulting in high coherence across the batch.

Student 1: " his absence, a queer throb of terror rent the air; a small fellow marred the narrow passage and turned his head round. Then he spoke, in a long, hollow voice, beseeching God. Then he repeated the word he had heard. Then the voice croaked: Who art thou,"

Score: 1.0

Student 2: " the fascatic silence of the night, a queer little figure passed the window and bid out a familiar, croaking noise underneath the window. old man paused; the words would have been echoed from his lips, and the words rang out in the sentry's ear: Stand still! Speak"

Score: 1.0

Student 3: " my house, a man's boy had paused, and presently he presented himself in sight with a white face like a stout man embracing a young woman's breast, and bending boldly in front, sang aloud, Go on, Master Chaunton: I pass thee without touching; I see thee without touching."

Score: 1.0

Student 4: " the afternoon a queer picture of the old man's little face appeared on the threshold of his hut; and he spoke in his ear: Your servant, my patient, has spoken at last. What's taken you so long? Speak! Do you know tell something and call me, or my white tongue"

Score: 1.0

DPO DPO relies on generating pairs and optimizing the model to prefer the higher-scoring output. Below are two examples in **step 26459** and **step 26537** showing the "Winner" vs. "Loser" dynamic during training.

Example 1:

Student 1 (loser): " whisper in my ear, I heard the voice of someone in the hall close to me: Who is it that I call this dead man by his name and speak of? It is my promise that you will not speak again on

my account; I shall not speak again for the whole of the night."

Score: 0.78

Student 2 (Winner): " whisper, a sound broke upon his ears: Tell me, Noel, what is the matter? Has there been a strange death in the neighbourhood? Is there not greater danger than that which has passed through my heart? Tell me, told you, what have you become of this curious stranger? What news"

Score: 1.0

Example 2:

Student 1 (Loser): " answer to her curtsey, a voice broke in upon the study: Poor old Sylvester, I have been almost certain of what is taken place inside this house, witch-girl, one last night and it is said to be a mystery. Can you see it?,"

Score: 0.78

Student 2 (Winner): " my restless, wistful voice, a voice cried out Refrain me Remains, heard it none the less; and I held it up, seeing it I had hidden it from my lips, and I knew, when I spoke Do say what I say, my"

Score: 0.89